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Project Title:	Modifying Protein-Lipid Interactions of a Microbial Mechanosensitive Channel: Functional Screens after Tethering Specific Residues to the Membrane		
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Task Description:	At the foundation of the ability of organisms to detect gravity is the capacity to detect mechanical forces. It is, for humans, the sheering force on stereocillia of vestibular hair cells, and for plants, the sedimentation of amyloplasts within a root cell. One of the primary molecules underlying the ability to detect mechanical forces is the mechanosensitive channel. To date, the best characterized is MscL, a microbial MechanoSensitive Channel of Large conductance. MscL opens a large pore in response to membrane tension; several studies now suggest sensing membrane tension is common among many mechanosensitive channels from microbes to plants and animals. Hence, the interactions between mechanosensitive channels and the lipid-membrane environment are of tremendous interest. Manipulating these interactions should have functional consequences that yield important clues of how the channel normally interacts with lipid. We have generated and characterized a library of mutated MscL channels in which each residue in and around the transmembrane domains has been sequentially mutated to cysteine. Treatment of these mutants with sulfhydryl reagents containing aromatic rings or alkyl chains should encourage hydrophobic interactions or tether the residue to the membrane. If the modified residue is normally within a hydrophobic environment primarily when in a specific state (closed, transition or open), then that state would be stabilized. We have developed an in vivo screen that allows us to rapidly assess functional changes in the MscL channel simply by monitoring bacterial growth; channels in which the open, closed or even various transition states have been stabilized are identified. Once manipulations that lead to functional consequences are selected, the exact functional changes of MscL will be assessed by electrophysiological assays. Preliminary experiments have demonstrated the feasibility of this approach. When completed, this study will yield critical insight into how mechanosensitive channels sense and respond to membrane tension.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many mechanosensors, most likely including those involved with gravity sensing, appear to sense and responds to biophysical properties of the lipid environment. Thus, we have been studying the protein-lipid interactions of a bacterial mechanosensor by using an in vivo screen (as described in detail in the proposal). We have found a “hot spot” near the second transmembrane domain, just proximal to a cytoplasmic cluster of charged amino acids. The data suggest that the residues just proximal to this cluster enter the lipid environment upon sensor stimulation. This finding may have implications for the functioning of other sensors: a cytoplasmic charge cluster is a common feature among many channels, including candidate mechanosensors. We speculate that the movement of this region into the membrane could be an underlying feature for the function of many mechanosensors.
Task Progress:	Progress: Underlying any organism’s ability to sense and react to gravitational force are biological mechano-sensors. In plants it is the sedimentation of amyloplasts within the columella cells in the root cap that generates the mechanical force; in humans it is the sheering forces along the stereocillia of the hair cells within the vestibular system. Although the precise molecular sensor in each of these cases is not yet known, most scientists within these fields believe that the molecular sensors are mechanosensitive (MS) ion channels. Recent studies have implied that many independent mechanosensors may have common molecular mechanisms, that of directly sensing their membrane environment. The bacterial mechanosensitive channel MscL (for MechanoSensitive Channel of Large conductance) is the first cloned and, to date, best characterized MS channel; it has emerged as a paradigm for how MS channels sense and respond to mechanical forces. The MscL channel opens a large pore in the cell envelope, which is composed of a lipid membrane, in response to acute decreases in osmotic environment, thus serving as a “biological emergency release valve”. Early studies assumed that the protein sensed tension within the cell lipid membrane; we have recently demonstrated that indeed this is the case. A plethora of evidence has suggested that a number of other independent MS channels similarly sense membrane stretch and either open in response to this stimulus, or have their opening modulated by it. The list of channels that sense membrane tension includes members of the bacterial and plant MscS MS channel family, a mechanosensitive channel from the yeast S. cerevisiae, and within mammals, members off the two-pore domain potassium (K2P), transient receptor potential (TRP), epithelial sodium (ENaC), and BK channels. Members within this list may indeed be related to gravity sensors used by plants (MscS homologues are among the best candidates) and animals (TRP channels are prime candidates). Thus, it appears that for many mechanosensors, most likely including those involved with gravity sensing, the protein senses and responds to its lipid environment. Sensor protein-lipid interactions therefore are likely to be important in sensing gravity. We hypothesized that modifying or exaggerating protein/lipid interactions would alter the normal response of a MS channel, and that these alterations would give insight into the fundamental principles of how a protein can sense and respond to membrane tension. Toward this end, we proposed the single objective of potentially modifying such interactions at over 75 sites within the 136 amino acid MscL protein. Briefly, we have generated and characterized a library of mutated MscL channels in which each residue in and around the membrane has been sequentially mutated to the reactive residue cysteine; note that MscL contains no endogenous cysteines. Treatment of these mutants with reactive chemicals, known as sulfhydryl reagents, containing hydrophobic aromatic rings or alkyl chains will tether the residue to the membrane or other hydrophobic pockets. If the residue is normally within these environments in a specific state (closed, open or transition), then that state would be stabilized. To generate rapid assays, we have used the power of microbial screens – we let the bacteria tell us what areas of the channel are functionally important. We have already developed a series of in vivo screens that allow us to assess functional changes in the MscL channel simply by monitoring bacterial growth and survival; in a single screen we can detect channels in which either the open, closed, and even transition states have been stabilized. Once potential mutants and hydrophobic- or lipid-tethering manipulations have been identified in a screen, the precise functional consequences to the channel properties will be assessed by conventional, but more laborious, electrophysiological assay known as patch clamp. Patch clamp allows scientists to measure the activities (opening and closing) of individual channels within a biological membrane; for the first time individual molecular events could be measured. It literally revolutionized the study of biological channels. Mechanosensitive (MS) channels that respond to membrane stretch were, for the first time, looked for and found in embryonic chick skeletal muscle in 1984. Since then, MS channels have been found in numerous cell types, and are likely to be the molecular sensors at the foundation of many senses including gravity sensing. In 1987 researchers generated giant bacterial cells and used patch clamp to assay channel activities within native E. coli membranes. What they found was an abundance of MS channels. We now know that there are at least two major families of MS channels within bacteria: MscL (for MechanoSensitive Channel of Large conductance) and MscS (Small conductance). MscL remains the best characterized MS channel to date. Reconstitution experiments have demonstrated that MscL

directly senses membrane tension, no other proteins are necessary for channel activity. Indeed, the channel appears to directly sense biophysical stresses within the lipid bilayer. Physiologically we know that MscL and MscS work in coordination as biological “emergency release valves” releasing excessive cell turgor (pressure within the cell) caused by acute decreases in osmotic environment, often referred to as osmotic downshock. Upon activation, a large pore is created that allows passage of molecules upwards of 30Å in diameter. Structural studies have solved a closed, or nearly closed, conformation of an MscL orthologue from Mycobacterium tuberculosis that reveals the complex as a homopentamer of subunits containing two transmembrane domains (TM1 and TM2) connected by a periplasmic loop; the amino- and carboxyl-termini are predicted to be cytoplasmic. The N-terminal forms an amphipathic (one side hydrophobic or lipid loving, the other hydrophilic or water loving) ? helix and is often referred to as S1. This structure is now known to run along the cytoplasmic face of the membrane where it may serve as a lipid anchor; this structure is shared among many channels and further understanding the importance of its interaction with the membrane may have far-reaching implications.

We have thus far performed the screen of the current cysteine library with the three compounds discussed in the proposal: 4-HB-MTS, which adds a tyrosine-like residue; MTSBn, which adds a phenylalanine-like residue; and DMTS, which adds a decyl fatty acid chain to the location. The percentage change in cell viability of the in vivo screen has been assessed. We have thus far obtained one publication in one of the more obvious ‘hot spots’. We have asked for a no-cost-extension for the remaining funds so we can pursue the following:

• We are investigating additional ‘hot spots’ within the protein by patch clamp

• We have extended the cysteine library to be screened to include the periplasmic loop between the two transmembrane domains. We are in the process of screening and investigating this region.

Bibliography Type:	Description: (Last Updated: 04/09/2020)
Articles in Peer-reviewed Journals	Iscla I, Wray R, Blount P. "An in vivo screen reveals protein-lipid interactions crucial for gating a mechanosensitive channel." FASEB J. 2011 Feb;25(2):694-702. Epub 2010 Nov 10. PubMed PMID: 21068398 , Feb-2011
Articles in Peer-reviewed Journals	Yang LM, Blount P. "Manipulating the permeation of charged compounds through the MscL nanovalve." FASEB J. 2011 Jan;25(1):428-34. Epub 2010 Oct 7. PubMed PMID: 20930114 , Jan-2011
Articles in Peer-reviewed Journals	Dorwart MR, Wray R, Brautigam CA, Jiang Y, Blount P. "S. aureus MscL is a pentamer in vivo but of variable stoichiometries in vitro: implications for detergent-solubilized membrane proteins." PLoS Biol. 2010 Dec 7;8(12):e1000555. PubMed PMID: 21151884 , Dec-2010